Answer **all** the questions.

1 Given that $\frac{8^{-1}}{8^k} \times 64 = 1$, find k.

Answer $k =$ .... [2]

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2 Find the difference between the greatest and the least integer values of x that satisfies $-4 \leq \frac{x-3}{4} - \frac{x+2}{3} < 2$.

Answer .... [3]

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3 **(a)** 12 engineers worked together and completed 30 identical model cars in 6 days. Assuming all the engineers worked at the same rate, how many days would 18 engineers take to complete 15 identical model cars?

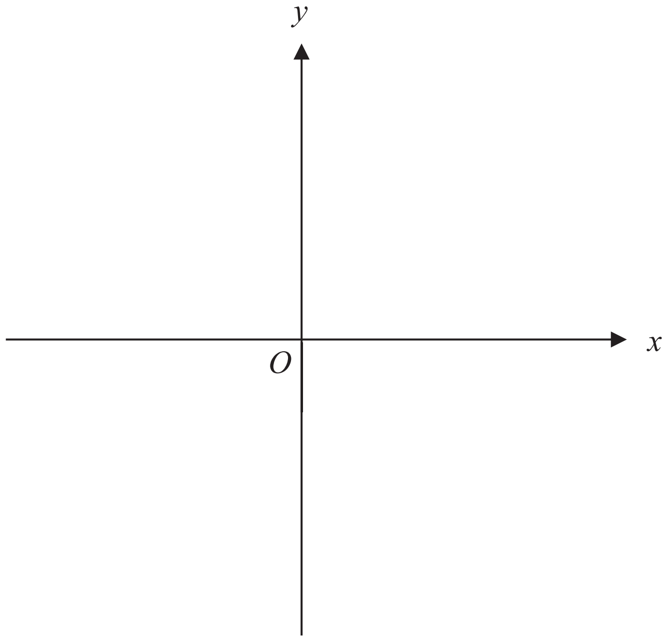
Answer .... days [2]

(b) The kinetic energy of a moving object, E joules, is directly proportional to the square of its speed, v m/s.
When the speed of the moving object is halved, the corresponding kinetic energy of the object is p times its original kinetic energy.

Find the value of p.

Answer $p =$ .... [2]

4 Sketch the graph of $y = 16 - (x-3)^2$ on the axes below.
Indicate clearly the coordinates of the points where the graph crosses the axes and the maximum point on the curve.



[2]